# COLLABORATING WITH GITHUB

CS 5004, SPRING 2026 – LECTURE 7

# NOW YOU'RE WORKING IN GROUPS…

**Good git hygiene is vital!**

Why? Merge conflicts ☹

# AVOIDING MERGE CONFLICTS

**Golden rule #1: Do not touch code/files that someone else is likely to edit**

If a teammate is working on SomeClass.java, DO NOT

- edit SomeClass.java
- move SomeClass.java
- rename/change the package containing SomeClass.java

Discuss who is working on what before touching anything!

# AVOIDING MERGE CONFLICTS

**Golden rule #2: ALWAYS start work with the following commands**

```
git status
<commit and push any outstanding local changes>
git pull
```

Even if you only left your computer for a short time and did not shut your computer/files e.g.

- Left the IDE open overnight
- Went to the store
- Made a sandwich

# AVOIDING MERGE CONFLICTS

**Golden rule #3: NEVER work directly on the MAIN branch**

When beginning a new assignment, create a new “topic branch” just for you (instructions on Canvas)

- Do all your work on that branch
- Merge to master only when your topic branch is working without compile time errors and is thoroughly tested

# IF A MERGE CONFLICT HAPPENS…

**1) Don’t panic**

**2) DO NOT use –f or --force**

No matter how many Stack Overflow posts tell you to do it!

# PRACTICE WITH BRANCHES

**In your own repo, using the command line (not the GUI or IntelliJ):**

`cd path/to/your/repo`

`git status` (commit and push any outstanding changes, if needed)

`git pull`

`git checkout -b branch_name`

Write some code, commit and push

→ you will need to set the remote branch on first push:

`git push --set-upstream origin branch_name`